

ASSESSMENT TOOL 1 – Application-Based Questions (3 Questions)

Course Code: ITA0609

Course Title: Machine Learning

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Question 1 (5 Marks)

A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. Find the probability that the car came from Section C.

Solution

This problem is solved using **Bayes' Theorem**, which is used to find the probability of an event based on given information.

Step 1: Define the Events

Let,

- A = Car is selected from Section A
- B = Car is selected from Section B
- C = Car is selected from Section C
- Blue = The selected car is blue

We need to find:

$P(C | \text{Blue})$

This represents the probability that the selected blue car came from Section C.

Step 2: Find the Prior Probabilities

The question states that **Section B is twice as likely to be selected as Sections A and C.**

Assume:

$$P(A) = x$$

$$P(B) = 2x$$

$$P(C) = x$$

Since the total probability must be 1,

$$x + 2x + x = 1$$

$$4x = 1$$

$$x = 1/4$$

Therefore,

$$P(A) = 1/4$$

$$P(B) = 1/2$$

$$P(C) = 1/4$$

Step 3: Find the Probability of Selecting a Blue Car from Each Section

Section A

Number of red cars = 5

Number of blue cars = 3

Number of black cars = 2

Total cars = $5 + 3 + 2 = 10$

Therefore,

$$P(\text{Blue} | A) = 3/10$$

Section B

Number of red cars = 4

Number of blue cars = 6

Number of black cars = 5

$$\text{Total cars} = 4 + 6 + 5 = 15$$

Therefore,

$$P(\text{Blue} | B) = 6/15 = 2/5$$

Section C

$$\text{Number of red cars} = 7$$

$$\text{Number of blue cars} = 2$$

$$\text{Number of black cars} = 6$$

$$\text{Total cars} = 7 + 2 + 6 = 15$$

Therefore,

$$P(\text{Blue} | C) = 2/15$$

Step 4: Apply Bayes' Theorem

Bayes' Theorem is

$$P(C | \text{Blue}) = [P(\text{Blue} | C) \times P(C)] / [P(\text{Blue} | A) \times P(A) + P(\text{Blue} | B) \times P(B) + P(\text{Blue} | C) \times P(C)]$$

Substitute the values:

$$P(C | \text{Blue})$$

$$= [(2/15) \times (1/4)] \div [(3/10 \times 1/4) + (2/5 \times 1/2) + (2/15 \times 1/4)]$$

Step 5: Calculate the Numerator

$$(2/15) \times (1/4)$$

$$= 2/60$$

$$= 1/30$$

Step 6: Calculate the Denominator

$$(3/10 \times 1/4) = 3/40$$

$$(2/5 \times 1/2) = 1/5$$

$$(2/15 \times 1/4) = 1/30$$

Taking the LCM as 120,

$$3/40 = 9/120$$

$$1/5 = 24/120$$

$$1/30 = 4/120$$

Therefore,

Denominator

$$= (9 + 24 + 4)/120$$

$$= 37/120$$

Step 7: Final Calculation

$$P(C \mid \text{Blue})$$

$$= (1/30) \div (37/120)$$

$$= (1/30) \times (120/37)$$

$$= 4/37$$

$$= 0.1081$$

Final Answer

Probability that the selected blue car came from Section C = $4/37$

Decimal Value = 0.1081

Percentage = 10.81%

Conclusion

Using **Bayes' Theorem**, the probability that the randomly selected blue car belongs to **Section C** is **$4/37$** , which is approximately **10.81%**.

Question 2 (5 Marks)

Examine the performance of a machine-learning model using its training and validation loss values over multiple epochs.

Given Dataset

Epoch	Training Loss	Validation Loss
1	2.8	2.4
2	2.5	2.2
3	2.2	2.0
4	2.0	2.1
5	1.8	2.3
6	1.6	2.6
7	1.5	2.9
8	1.3	3.2

Solution:

i) Identify the epoch at which the model begins to overfit.

To identify overfitting, compare the **training loss** and **validation loss** at each epoch.

- From **Epoch 1 to Epoch 3**, both training loss and validation loss decrease. This indicates that the model is learning effectively and improving its performance.
- At **Epoch 4**, the training loss continues to decrease ($2.2 \rightarrow 2.0$), but the validation loss increases ($2.0 \rightarrow 2.1$).
- From **Epoch 5 onwards**, the training loss keeps decreasing while the validation loss continues to increase significantly.

This indicates that the model is memorizing the training data instead of learning general patterns.

Therefore, the model begins to overfit at Epoch 4.

ii) Suggest two effective techniques to reduce overfitting.

1. Early Stopping

Early stopping monitors the validation loss during training. When the validation loss starts increasing while the training loss continues to decrease, training is stopped automatically.

Benefit: Prevents the model from learning unnecessary details from the training data.

2. Regularization (L1/L2 or Dropout)

Regularization adds a penalty to the loss function or randomly disables some neurons during training (Dropout).

Benefit: Reduces model complexity and prevents the model from memorizing the training data, improving its performance on unseen data.

iii) Discuss how overfitting affects the model's generalization ability on unseen data.

Overfitting occurs when a machine learning model learns the training data too closely, including noise and unnecessary patterns.

As a result:

- The model achieves very low training loss.
- The validation loss increases because the model cannot correctly predict new data.
- The model performs well on the training dataset but poorly on unseen or real-world data.
- The overall accuracy and reliability of the model decrease.

Therefore, overfitting reduces the model's ability to generalize and makes it less effective for practical applications.

Final Answer

i)

The model begins to **overfit at Epoch 4**, where the training loss decreases but the validation loss starts increasing.

ii)

Two techniques to reduce overfitting are:

- Early Stopping
- Regularization (L1/L2) or Dropout

iii)

Overfitting reduces the model's ability to generalize because it memorizes the training data instead of learning meaningful patterns. As a result, it performs well on training data but poorly on unseen data, leading to lower prediction accuracy.

Question 3 (5 Marks)

Examine the performance of a machine-learning model using training and validation accuracy over multiple epochs.

Given Dataset

Epoch	Training Accuracy	Validation Accuracy
1	60%	58%
2	65%	63%
3	70%	68%
4	75%	72%
5	80%	73%
6	85%	72%
7	88%	70%
8	92%	68%

Solution

i) Identify the epoch at which the model starts overfitting.

Overfitting occurs when the **training accuracy continues to increase**, while the **validation accuracy stops improving or starts decreasing**.

From the given data:

- **Epochs 1 to 5:** Both training accuracy and validation accuracy increase, indicating that the model is learning useful patterns.
- **Epoch 6:** Training accuracy increases from **80% to 85%**, but validation accuracy decreases from **73% to 72%**.
- **Epochs 7 and 8:** Training accuracy continues to improve, while validation accuracy decreases further.

Therefore, **the model starts overfitting at Epoch 6**.

ii) Explain why validation accuracy decreases despite training accuracy increasing.

Validation accuracy decreases because the model begins to **overfit** the training data.

During overfitting:

- The model memorizes the training examples instead of learning general patterns.
- Training accuracy continues to improve because the model becomes highly specialized for the training dataset.
- However, the model cannot perform well on new or unseen validation data.
- As a result, the validation accuracy decreases even though the training accuracy increases.

Thus, the model loses its ability to generalize to unseen data.

iii) Suggest two methods to improve validation performance.

1. Early Stopping

Early stopping monitors the validation accuracy during training. When the validation accuracy stops improving or begins to decrease, training is stopped.

Benefit: Prevents the model from overfitting and improves its performance on unseen data.

2. Regularization (L2 Regularization or Dropout)

Regularization reduces the complexity of the model by adding a penalty to large weights or randomly disabling neurons during training (Dropout).

Benefit: Helps the model learn general patterns instead of memorizing the training data, resulting in better validation performance.

Final Answer

i)

The model starts **overfitting at Epoch 6**, because the validation accuracy begins to decrease while the training accuracy continues to increase.

ii)

Validation accuracy decreases because the model starts memorizing the training data instead of learning general features. This reduces its ability to perform well on unseen validation data.

iii)

Two methods to improve validation performance are:

- **Early Stopping** – Stops training when validation accuracy begins to decrease.
- **Regularization (L2 Regularization or Dropout)** – Reduces overfitting by preventing the model from memorizing the training data.

